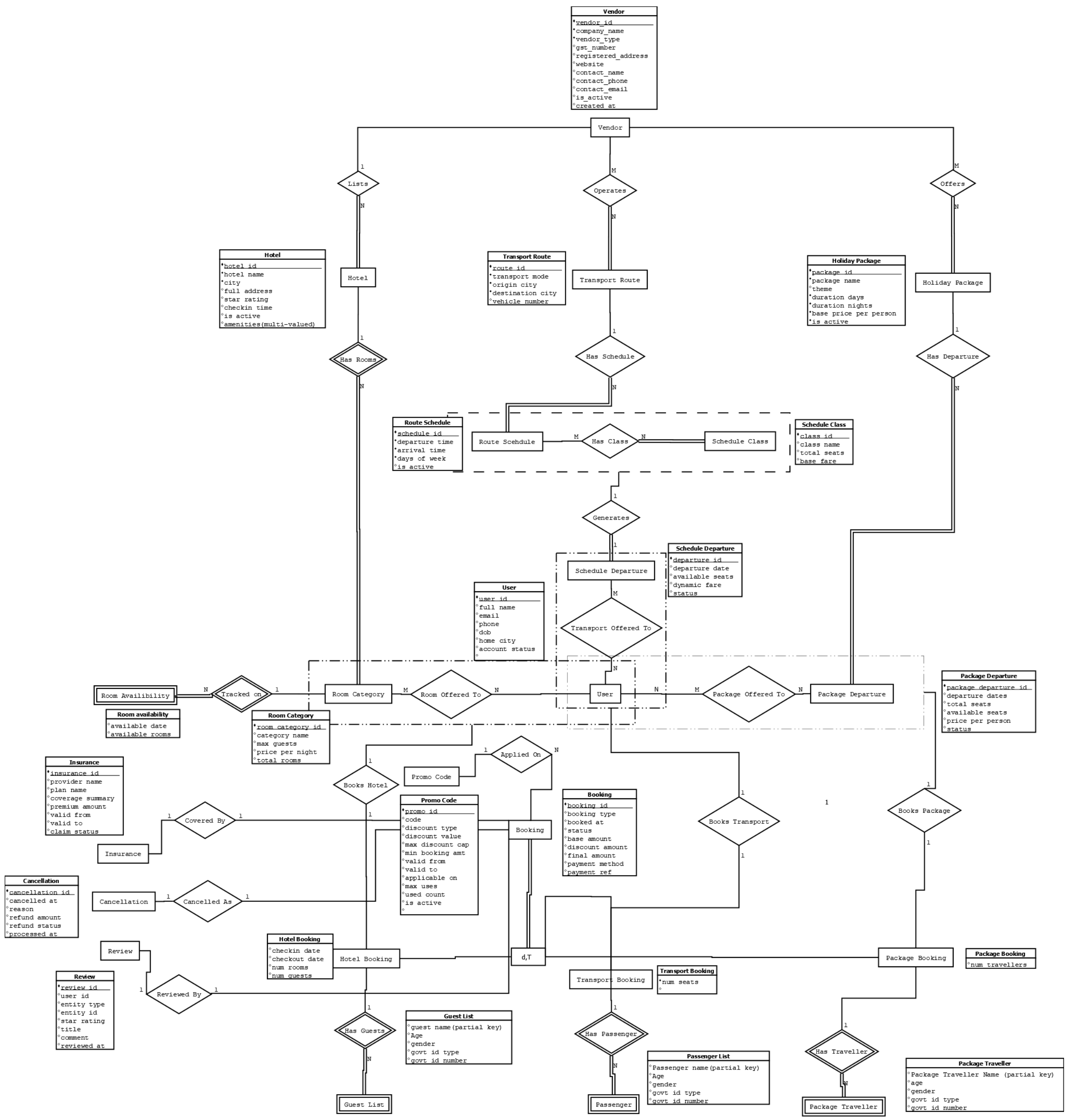
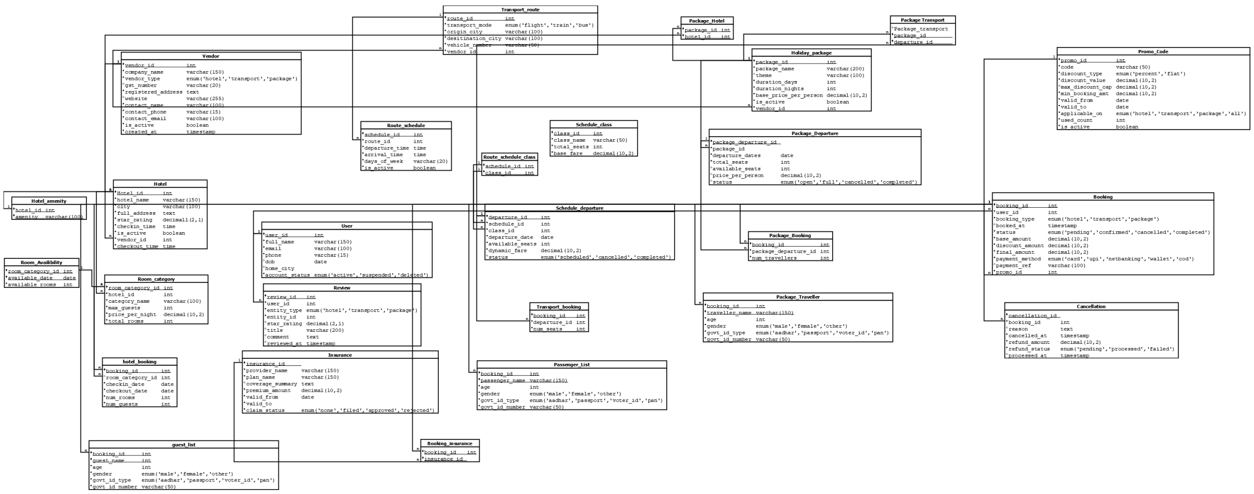
# ERD, Relational Schema and Normalization Proofs

## ER Diagram



# Relational Schema



# Normalization Proofs

## Core Entities

- Vendor
- Hotel
- Hotel_Amenity
- Room_Category
- Room_Availability
- User
- Review
- Insurance

## Transport

- Transport_Route
- Route_Schedule
- Schedule_Class

- Route_Schedule_Class
- Schedule_Departure

## Packages

- Holiday_Package
- Package_Departure

## Bookings

- Promo_Code
- Booking
- Hotel_Booking
- Guest_List
- Transport_Booking
- Passenger_List
- Package_Booking
- Package_Traveller
- Cancellation

## Supporting

- Booking_Insurance
- Package_Hotel
- Package_Transport

## Core Entities

## Vendor

ATTRIBUTES: vendor id, company_name, vendor_type, gst_number, registered_address, website, contact_name, contact_phone, contact_email, is_active, created_at

CANDIDATE KEY(S): {vendor_id}

## MINIMAL FD SET (Canonical Cover):

vendor_id → company_name, vendor_type, gst_number, registered_address, website,
contact_name, contact_phone, contact_email, is_active, created_at

## BCNF PROOF:

Candidate key: {vendor_id}.
The minimal FD set contains exactly one FD. Its determinant vendor_id is the sole candidate key, hence a superkey. Since every non-trivial FD has a superkey as its determinant, Vendor is in BCNF.

## Hotel

ATTRIBUTES: hotel id, hotel_name, city, full_address, star_rating, checkin_time, is_active, vendor_id

## CANDIDATE KEY(S): {hotel_id}

## MINIMAL FD SET (Canonical Cover):

hotel_id → hotel_name, city, full_address, star_rating, checkin_time, is_active, vendor_id

## BCNF PROOF:

Candidate key: {hotel_id}.
The single non-trivial FD has hotel_id - the sole candidate key - as its determinant, making it a superkey. No FD in the minimal set violates BCNF. Hotel is in BCNF.

Note: the multi-valued attribute ‘amenities’ is factored out into Hotel_Amenity.

## Hotel_Amenity

ATTRIBUTES: hotel id, amenity

## CANDIDATE KEY(S): {(hotel_id, amenity)}

## MINIMAL FD SET (Canonical Cover):

(No non-trivial FDs - the key spans the entire relation)

## BCNF PROOF:

Candidate key: {hotel_id, amenity}.
The relation has only two attributes, both forming the key. There are no non-trivial FDs, so the condition for BCNF is satisfied vacuously. Hotel_Amenity is in BCNF.

## Room_Category

ATTRIBUTES: room category id, hotel_id, category_name, max_guests, price_per_night, total_rooms
CANDIDATE KEY(S): {room_category_id}

## MINIMAL FD SET (Canonical Cover):

room_category_id → hotel_id, category_name, max_guests, price_per_night, total_rooms

## BCNF PROOF:

Candidate key: {room_category_id}.
The single FD's determinant is room_category_id, the sole candidate key and therefore a superkey. Room_Category is in BCNF.

## Room_Availability

ATTRIBUTES: room category id, available date, available_rooms

## CANDIDATE KEY(S): {(room_category_id, available_date)}

## MINIMAL FD SET (Canonical Cover):

room_category_id, available_date → available_rooms

## BCNF PROOF:

Candidate key: {room_category_id, available_date}.
The composite key is the only determinant that appears in the minimal FD set. A composite key is a superkey, so Room_Availability is in BCNF.

## User

ATTRIBUTES: user id, full_name, email, phone, dob, home_city, account_status

## CANDIDATE KEY(S): {user_id} , {email}

## MINIMAL FD SET (Canonical Cover):

user_id → full_name, email, phone, dob, home_city, account_status
email → user_id

## BCNF PROOF:

Candidate keys: {user_id} and {email} (email is unique per user).
FD 1: determinant user_id is a candidate key (superkey).
FD 2: determinant email is a candidate key (superkey).
Both FDs in the minimal set have superkeys as determinants. User is in BCNF.

## Review

ATTRIBUTES: review id, user_id, entity_type, entity_id, star_rating, title, comment, reviewed_at
CANDIDATE KEY(S): {review_id}

## MINIMAL FD SET (Canonical Cover):

review_id → user_id, entity_type, entity_id, star_rating, title, comment, reviewed_at

## BCNF PROOF:

Candidate key: {review_id}.
The determinant review_id is the sole candidate key, hence a superkey. Review is in BCNF.

## Insurance

ATTRIBUTES: insurance id, provider_name, plan_name, coverage_summary, premium_amount, valid_from, valid_to, claim_status

CANDIDATE KEY(S): {insurance_id}

MINIMAL FD SET (Canonical Cover):
insurance_id → provider_name, plan_name, coverage_summary, premium_amount, valid_from, valid_to, claim_status

## BCNF PROOF:

Candidate key: {insurance_id}.
The single FD's determinant is insurance_id, the sole candidate key and a superkey. Insurance is in BCNF.

## Transport

## Transport_Route

ATTRIBUTES: route id, transport_mode, origin_city, destination_city, vehicle_number, vendor_id
CANDIDATE KEY(S): {route_id}

MINIMAL FD SET (Canonical Cover):
route_id → transport_mode, origin_city, destination_city, vehicle_number, vendor_id

## BCNF PROOF:

Candidate key: {route_id}.
route_id is the sole candidate key, so it is a superkey. Transport_Route is in BCNF.

## Route_Schedule

ATTRIBUTES: schedule id, route_id, departure_time, arrival_time, days_of_week, is_active
CANDIDATE KEY(S): {schedule_id}

MINIMAL FD SET (Canonical Cover):
schedule_id → route_id, departure_time, arrival_time, days_of_week, is_active

## BCNF PROOF:

Candidate key: {schedule_id}.
schedule_id is a superkey. Route_Schedule is in BCNF.

## Schedule_Class

ATTRIBUTES: class id, class_name, total_seats, base_fare
CANDIDATE KEY(S): {class_id}

MINIMAL FD SET (Canonical Cover):
class_id → class_name, total_seats, base_fare

## BCNF PROOF:

Candidate key: {class_id}.
class_id is the sole candidate key and a superkey. Schedule_Class is in BCNF.

## Route_Schedule_Class

ATTRIBUTES: schedule id, class id
CANDIDATE KEY(S): {(schedule_id, class_id)}

## MINIMAL FD SET (Canonical Cover):

(No non-trivial FDs - the key spans the entire relation)

## BCNF PROOF:

Candidate key: {schedule_id, class_id}.
This is a pure junction relation. The key covers all attributes; no proper-subset FDs

exist. BCNF holds vacuously.
Route_Schedule_Class is in BCNF.

## Schedule_Departure

ATTRIBUTES: departure id, schedule_id, class_id, departure_date, available_seats, dynamic_fare, status
CANDIDATE KEY(S): {departure_id}

MINIMAL FD SET (Canonical Cover):
departure_id → schedule_id, class_id, departure_date, available_seats, dynamic_fare, status

## BCNF PROOF:

Candidate key: {departure_id}.
departure_id is the sole candidate key, hence a superkey. Schedule_Departure is in BCNF.

## Packages

## Holiday_Package

ATTRIBUTES: package id, package_name, theme, duration_days, duration_nights, base_price_per_person, is_active, vendor_id

CANDIDATE KEY(S): {package_id}

MINIMAL FD SET (Canonical Cover):

package_id → package_name, theme, duration_days, duration_nights, base_price_per_person, is_active, vendor_id

## BCNF PROOF:

Candidate key: {package_id}.
package_id is a superkey. Holiday_Package is in BCNF.

## Package_Departure

ATTRIBUTES: package departure id, package_id, departure_dates, total_seats, available_seats, price_per_person, status

CANDIDATE KEY(S): {package_departure_id}

MINIMAL FD SET (Canonical Cover):
package_departure_id → package_id, departure_dates, total_seats, available_seats, price_per_person, status

## BCNF PROOF:

Candidate key: {package_departure_id}.
package_departure_id is a superkey. Package_Departure is in BCNF.

# Bookings

## Promo_Code

ATTRIBUTES: promo id, code, discount_type, discount_value, max_discount_cap, min_booking_amt, valid_from, valid_to, applicable_on, used_count, is_active

CANDIDATE KEY(S): {promo_id}, {code}}

## MINIMAL FD SET (Canonical Cover):

promo_id → code, discount_type, discount_value, max_discount_cap,
min_booking_amt,
valid_from, valid_to, applicable_on, used_count, is_active
code → promo_id

## BCNF PROOF:

Candidate keys: {promo_id} and {code} (promo codes are unique).
FD 1: determinant promo_id is a candidate key (superkey).
FD 2: determinant code is a candidate key (superkey).
Both determinants are superkeys. Promo_Code is in BCNF.

## Booking

ATTRIBUTES: booking id, user_id, booking_type, booked_at, status, base_amount, discount_amount, final_amount, payment_method, payment_ref, promo_id

## CANDIDATE KEY(S): {booking_id}

## MINIMAL FD SET (Canonical Cover):

booking_id → user_id, booking_type, booked_at, status, base_amount, discount_amount,
final_amount, payment_method, payment_ref, promo_id

## BCNF PROOF:

Candidate key: {booking_id}.
booking_id is the sole candidate key, hence a superkey. Booking is in BCNF.

## Hotel_Booking

BCNF

ATTRIBUTES: booking id, room_category_id, checkin_date, checkout_date, num_rooms, num_guests

## CANDIDATE KEY(S): {booking_id}

## MINIMAL FD SET (Canonical Cover):

booking_id → room_category_id, checkin_date, checkout_date, num_rooms, num_guests

## BCNF PROOF:

Candidate key: {booking_id} (1-to-1 specialisation of Booking).
booking_id is the sole candidate key and a superkey. Hotel_Booking is in BCNF.

## Guest_List

ATTRIBUTES: booking id, quest name, age, gender, govt_id_type, govt_id_number
CANDIDATE KEY(S): {(booking_id, guest_name)}

MINIMAL FD SET (Canonical Cover):
booking_id, guest_name → age, gender, govt_id_type, govt_id_number

## BCNF PROOF:

Candidate key: {booking_id, guest_name} (weak entity identified within a booking).
The composite key is a superkey. The single FD's determinant equals the candidate key. Guest_List is in BCNF.

## Transport_Booking

ATTRIBUTES: booking id, departure_id, num_seats
CANDIDATE KEY(S): {booking_id}

MINIMAL FD SET (Canonical Cover):
booking_id → departure_id, num_seats

## BCNF PROOF:

Candidate key: {booking_id} (1-to-1 specialisation of Booking).
booking_id is a superkey. Transport_Booking is in BCNF.

## Passenger_List

ATTRIBUTES: booking id, passenger name, age, gender, govt_id_type, govt_id_number
CANDIDATE KEY(S): {(booking_id, passenger_name)}

MINIMAL FD SET (Canonical Cover):
booking_id, passenger_name → age, gender, govt_id_type, govt_id_number

## BCNF PROOF:

Candidate key: {booking_id, passenger_name} (weak entity).
The composite candidate key is a superkey. Passenger_List is in BCNF.

## Package_Booking

BCNF

ATTRIBUTES: booking id, package_departure_id, num_travellers
CANDIDATE KEY(S): {booking_id}

MINIMAL FD SET (Canonical Cover):
booking_id → package_departure_id, num_travellers

## BCNF PROOF:

Candidate key: {booking_id} (1-to-1 specialisation of Booking).
booking_id is a superkey. Package_Booking is in BCNF.

## Package_Traveller

ATTRIBUTES: booking id, traveller name, age, gender, govt_id_type, govt_id_number
CANDIDATE KEY(S): {(booking_id, traveller_name)}

MINIMAL FD SET (Canonical Cover):
booking_id, traveller_name → age, gender, govt_id_type, govt_id_number

## BCNF PROOF:

Candidate key: {booking_id, traveller_name} (weak entity).
The composite candidate key is a superkey. Package_Traveller is in BCNF.

## Cancellation

ATTRIBUTES: cancellation id, booking_id, reason, cancelled_at, refund_amount, refund_status, processed_at CANDIDATE KEY(S): {cancellation_id}

MINIMAL FD SET (Canonical Cover):
cancellation_id → booking_id, reason, cancelled_at, refund_amount, refund_status, processed_at

## BCNF PROOF:

Candidate key: {cancellation_id}.
cancellation_id is a superkey. Cancellation is in BCNF.

## Supporting

## Booking_Insurance

ATTRIBUTES: booking id, insurance id
CANDIDATE KEY(S): {(booking_id, insurance_id)}

MINIMAL FD SET (Canonical Cover):
(No non-trivial FDs - the key spans the entire relation)

## BCNF PROOF:

Candidate key: {booking_id, insurance_id}.
Pure junction relation. No proper-subset FDs exist. BCNF holds by default.

## Package_Hotel

ATTRIBUTES: package id, hotel id
CANDIDATE KEY(S): {(package_id, hotel_id)}

MINIMAL FD SET (Canonical Cover):
(No non-trivial FDs - the key spans the entire relation)
BCNF PROOF:
Candidate key: {package_id, hotel_id}.
Pure junction relation. BCNF holds by default.

## Package_Transport

ATTRIBUTES: package id, departure id
CANDIDATE KEY(S): {(package_id, departure_id)}

MINIMAL FD SET (Canonical Cover):
(No non-trivial FDs - the key spans the entire relation)

## BCNF PROOF:

Candidate key: {package_id, departure_id}.
Pure junction relation. BCNF holds by default.